

Clinical Risk Assessment Report: Patient 7

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.41x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.1%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -2.05 [Bottom 3%] (-2.1%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 1.13 [Top 2%] (-1.2%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -1.26 [Bottom 10%] (-1.0%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -4.59 [Bottom <1%] (-0.9%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.75 [Bottom 7%] (-0.8%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.41 [Bottom <1%] (-0.7%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.55 [Top 2%] (-0.6%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: -3.52 [Bottom <1%] (-0.5%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Machine Learning Feature Attribution (SHAP Decision Path)

